Abiodun Modupe, Ph.D.

Computer Science • Artificial Intelligence • Cybersecurity • Natural Language Processing
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U.S. Permanent Resident — authorized to work in the United States without employer sponsorship

Research Interests

Trustworthy and responsible artificial intelligence; multilingual and low-resource natural language processing; AI for cybersecurity and digital forensics; misinformation detection; explainability and evaluation of AI systems; and the security and reliability of agentic AI systems.

Teaching Areas

Artificial intelligence, machine learning, deep learning, natural language processing, data mining, computer security, operating systems, software engineering, programming, and capstone/project-based courses. Experience includes face-to-face, distance, hybrid, and fully online teaching.

Education

Ph.D., Computer Science **2024**
University of the Witwatersrand, South Africa
- Thesis: "Regularized Deep Neural Network for Post-Authorship Attribution." Supervisors: Prof. Turgay Celik and Prof. Vukosi Marivate.

Master of IT (Computer Science) **2011**
Tshwane University of Technology, South Africa
- Dissertation: "Intelligent Identification of Advanced Fee Fraud Activities on the Internet." Supervisors: Prof. O. O. Olugbara and Prof. S. O. Ojo.

B.Sc. (Hons), Computer Science **1997**
Obafemi Awolowo University, Nigeria

Academic Appointments

Lecturer & MIT Big Data Science Program Coordinator **2019–present**
Department of Computer Science, University of Pretoria
- Teaching remotely from the United States since 2026 while continuing program coordination and student advising.

Research Fellow **2021–present**
Data Science for Social Impact (DSFSI), University of Pretoria

Visiting Researcher **2018–2019**
Council for Scientific and Industrial Research (CSIR), Pretoria

Teaching Assistant **2017–2018**
University of the Witwatersrand

Independent Contractor / Distance-Learning Teaching Assistant **2013–2017**
University of South Africa

Lecturer (Contract) **2011–2017**
Tshwane University of Technology

Graduate Research/Teaching Assistant **2009–2011**
Tshwane University of Technology

Fellowships and Research Affiliations

Responsible AI Fellow **2024–present**
Stimson Center

AfriDSAI Fellow **2025–present**
University of Pretoria

Teaching and Program Leadership

University of Pretoria – Lecturer (Honours and Master's)

- COS760 Natural Language Processing (Honours), since 2025; co-taught.
- COS781 Data Mining (Honours), since 2024.
- COS801 Deep Learning (postgraduate elective; in-person and online), since 2022.
- MIT801 Introduction to Machine and Statistical Learning (co-taught; hybrid in 2026), since 2020.
- MIT808 Data Science Capstone Project (co-taught), since 2020; partner-led projects, peer assessment, poster exhibition, and technical reporting.
- MIT807 Mini-Dissertation in Big Data Science (90 credits), module coordinator, since 2020.

Earlier Teaching

- **Tshwane University of Technology:** ISC216D Information Security; ORS216D Operating Systems; RRS108G Research Project (Lecturer, 2011–2017).
- **Tshwane University of Technology:** DCT115D Discrete Structures; PPB115D Principles of Programming; AOP216D Advanced Object-Oriented Programming (Teaching Assistant, 2009–2011).
- **University of the Witwatersrand:** COMS7047A Adaptive Computation and Machine Learning (Teaching Assistant, 2018).
- **University of South Africa:** COS2621 Computer Organization, distance learning (Teaching Assistant/e-tutor, 2013–2017).
- **Deep Learning Indaba:** Tutor, 2017–2019 and 2023.

Program Coordination and Curriculum Leadership

- Coordinate the part-time MIT Big Data Science program, including admissions, orientation, academic progression, timetabling, capstone and mini-dissertation processes, and cross-departmental supervision.
- Advise annual cohorts of approximately 45–50 students and coordinate research milestones and examination processes for MIT807 mini-dissertations.

Student Supervision and Advising

- Completed supervision: 10 MIT Big Data Science mini-dissertations, 4 M.Sc. (Computer Science) dissertations, and 1 Ph.D. student.
- Current doctoral supervision and co-supervision spans Computer Science and Informatics at the University of Pretoria and North-West University.
- Several peer-reviewed publications listed below are first-authored by students I supervised or co-supervised.

Research Funding and Support

AI4D African Languages Lab – IDRC/FCDO **2024–present**
Principal Investigator (with Vukosi Marivate); CAD 1,070,500
- African-language NLP data and evaluation, capacity building, and work on equitable access, licensing, and policy.

AfriGemma – Google Research Fund **2025**
Co-PI; USD 200,000
- Adapting open Gemma models to African languages; open data, model, and tooling outputs.

Transaction Capital – Vehicle Inspection **2022–2024**
Research project with M.Sc. sponsorship; USD 60,000

Google AI Research – Cross-Lingual Modelling **2022–2023**
Research support with M.Sc. sponsorship; USD 25,000

Google exploreCSR Program **2023**
USD 18,000

Google Cloud Research Credits **2022**
USD 10,000

Mozilla Open-Source Support Award **2021**
Co-PI share USD 3,000 of USD 10,000

AWS Research Credits **2018**
USD 5,000

Peer-Reviewed Publications

Journal Articles

1. T. Sindane, V. Marivate, **A. Modupe**, and H. Combrink, "Transformers for sign language processing: A survey of spatio-temporal modeling approaches, challenges, and research gaps," *Intelligent Systems with Applications*, vol. 31, Art. no. 200723, 2026. doi:10.1016/j.iswa.2026.200723.
2. T. Sindane, V. Marivate, and **A. Modupe**, "Cross-lingual embedding methods and applications: A systematic review for low-resourced scenarios," *Natural Language Processing Journal*, vol. 12, pp. 1–20, 2025. doi:10.1016/j.nlp.2025.100157.
3. V. Marivate, I. J. Dzingirai, F. Banda, R. Lastrucci, T. Sindane, K. Madumo, K. Olaleye, **A. Modupe**, U. Netshifhefhe, H. Combrink, and M. Nakeng, "Mafoko: Structuring and Building Open Multilingual Terminologies for South African NLP," *Journal of the Digital Humanities Association of Southern Africa*, vol. 6, no. 2, 2025.
4. G. Shoderu, S. O. Baror, S. Makura, and **A. Modupe**, "Digital Forensic Readiness to Mitigate Insider Threats in the SaaS Cloud Environment," *The Indonesian Journal of Computer Science*, vol. 14, no. 6, pp. 10287–10300, 2025. doi:10.33022/ijcs.v14i6.5059.
5. S. Rananga, B. Isong, **A. Modupe**, and V. Marivate, "Misinformation Detection: A Review for High and Low-Resource Languages," *Journal of Information Systems and Informatics*, vol. 6, no. 4, pp. 2892–2922, 2024. doi:10.51519/journalisi.v6i4.931.
6. M. Mathebula, **A. Modupe**, and V. Marivate, "Fine-Tuning Retrieval-Augmented Generation with an Auto-Regressive Language Model for Sentiment Analysis in Financial Reviews," *Applied Sciences*, vol. 14, no. 23, p. 10782, 2024. doi:10.3390/app142310782.
7. M. Mathebula, **A. Modupe**, and V. Marivate, "ChatGPT as a Text Annotation Tool to Evaluate Sentiment Analysis on South African Financial Institutions," *IEEE Access*, vol. 12, pp. 144017–144043, 2024. doi:10.1109/ACCESS.2024.3464374.
8. **A. Modupe**, T. Celik, V. Marivate, and O. O. Olugbara, "Post-Authorship Attribution Using Regularized Deep Neural Network," *Applied Sciences*, vol. 12, no. 15, p. 7518, 2022. doi:10.3390/app12157518.

Manuscript Under Review

1. **A. Modupe** et al., "Anisomorphism-Aware Cross-Lingual Alignment for Low-Resource African Languages: A Causal Geometric Approach," *IEEE Access*, under review (resubmitted October 2026).

Conference Proceedings

1. I. Abdulmumin, T. Gwadabe, S. H. Muhammad, D. I. Adelani, N. Khalo, I. S. Ahmad, **A. Modupe** et al., "AfriScience-MT: Towards Decolonizing Science in Africa through Text Translation," *Proceedings of EMNLP 2026*, Budapest, Hungary, pp. 17506–17555.
2. **A. Modupe** et al., "Responsible AI in the Global South: An Ontology-Based Approach with Automated Policy Acquisition and NLP-Driven Dimension Scoring," 7th Southern African Conference for Artificial Intelligence Research (SACAIR 2026), accepted for Springer CCIS.
3. G. Shoderu, S. Baror, **A. Modupe**, and S. Makura, "Digital Forensic Readiness to Mitigate Insider Threats in the SaaS Cloud Environment," *Proceedings of the 21st International Conference on Cyber Warfare and Security (ICCWS 2026)*, pp. 427–437. doi:10.34190/iccws.21.1.4508.
4. A. Tegene, V. Marivate, M. Banda, **A. Modupe**, V. Rakotonarivo, and M. Nchabeleng, "An End-to-End Deep Learning Model for Recommender Systems," SACAIR 2025, Springer Nature, pp. 64–81.
5. C. du Plessis, M. du Plessis, K. R. Mabokela, **A. Modupe**, and V. Marivate, "Identification of Social Media Users that Perpetuate Xenophobic Attitudes and Hate Speech Narratives in South Africa," SACAIR, Springer, pp. 254–271, 2025. doi:10.1007/978-3-032-11733-5_16.
6. **A. Modupe**, T. Celik, V. Marivate, and O. O. Olugbara, "Integrating Bidirectional Long Short-Term Memory with Subword Embedding for Authorship Attribution," 2023 IEEE International Conference on Systems, Man, and Cybernetics (SMC), pp. 1910–1917. doi:10.1109/SMC53992.2023.10393898.
7. J. S. Dlamini, L. Marshall, and **A. Modupe**, "Financial Inclusion in Sub-Saharan Emerging Markets: The Application of Deep Learning to Improve Determinants," SACAIR, Springer, pp. 219–236, 2023.
8. A. De Jager, V. Marivate, and **A. Modupe**, "Multimodal Misinformation Detection in a South African Social Media Environment," SACAIR, Springer, pp. 285–299, 2023. doi:10.1007/978-3-031-49002-6_19.
9. L. Nthimo, T. Mokoena, **A. Modupe**, and V. Marivate, "Call Centre Shift Schedule Optimisation using Local Search Heuristics," IEEE AFRICON, pp. 1–6, 2021. doi:10.1109/AFRICON51333.2021.9570947.
10. M. Lusinga, T. Mokoena, **A. Modupe**, and V. Marivate, "Investigating Statistical and Machine Learning Techniques to Improve the Credit Approval Process in Developing Countries," IEEE AFRICON, pp. 1–6, 2021. doi:10.1109/AFRICON51333.2021.9571004.
11. V. Marivate, T. Sefara, V. Chabalala, K. Makhaya, T. Mokgonyane, R. Mokoena, and **A. Modupe**, "Investigating an Approach for Low Resource Language Dataset Creation, Curation and Classification: Setswana and Sepedi," First Workshop on Resources for African Indigenous Languages (RAIL), pp. 15–20, 2020.
12. T. J. Sefara and **A. Modupe**, "Yorùbá Gender Recognition from Speech Using Neural Networks," 6th International Conference on Soft Computing and Machine Intelligence (ISCMI), IEEE, pp. 50–55, 2019. doi:10.1109/ISCMI47871.2019.9004376.
13. **A. Modupe**, T. Celik, V. Marivate, and M. Diale, "Semi-Supervised Probabilistic Approach for Normalising Informal Short Text Messages," Conference on Information Communication Technology and Society (ICTAS), IEEE, pp. 1–8, 2017. doi: 10.1109/ICTAS.2017.7920659.
14. M. Diale, C. van der Walt, T. Celik, and **A. Modupe**, "Feature Selection and Support Vector Machine Hyper-Parameter Optimisation for Spam Detection," PRASA-RobMech, IEEE, pp. 1–7, 2016. doi:10.1109/RoboMech.2016.7813162.
15. A. A. Adedamola, **A. Modupe**, and O. J. Dehinbo, "Development and Evaluation of a System for Normalizing Internet Slangs in Social Media Texts," World Congress on Engineering and Computer Science (WCECS), vol. 1, IAENG, San Francisco, 2015.
16. I. A. Modupe, O. O. Olugbara, S. O. Ojo, and **A. Modupe**, "Experimental Comparison of Genetic Algorithm and Ant Colony Optimisation to Minimise Energy in Ad-hoc Wireless Networks," WCECS, vol. 2, IAENG, San Francisco, 2013.
17. I. A. Modupe, O. O. Olugbara, and **A. Modupe**, "Minimizing Energy Consumption in Wireless Ad Hoc Networks with Metaheuristics," *Procedia Computer Science*, vol. 19, pp. 106–115, 2013. doi:10.1016/j.procs.2013.06.019.
18. **A. Modupe**, O. O. Olugbara, and S. O. Ojo, "Investigating Topic Models for Mobile Short Messaging Service Communication Filtering," World Congress on Engineering (WCE), vol. 2, IAENG, London, 2013.
19. **A. Modupe**, O. O. Olugbara, and S. O. Ojo, "Comparing Supervised Learning Classifiers to Detect Advanced Fee Fraud Activities on Internet," in *Advances in Computer Science and Information Technology*, Springer, pp. 87–100, 2012. doi: 10.1007/978-3-642-27317-9_10.
20. **A. Modupe**, O. O. Olugbara, and S. O. Ojo, "Identifying Advanced Fee Fraud Activities on the Internet Using Machine Learning Algorithms," 3rd IEEE International Conference on Computational Intelligence and Industrial Application (PACIIA), pp. 240–242, 2010. doi:10.1109/PACIIA.2010.5495084.

Book Chapter

1. **A. Modupe**, O. O. Olugbara, and S. O. Ojo, "Filtering of Mobile Short Messaging Service Communication Using Latent Dirichlet Allocation with Social Network Analysis," in *Transactions on Engineering Technologies*, Springer, 2014. doi: 10.1007/978-94-017-8832-8_48.

Workshop Papers

1. R. Myoya, L. Matojane, V. Marivate, **A. Modupe**, and A. Myburgh, "Assessing the Feasibility of Identifying Unique Features of Wildlife from a Drone's Perspective," Fourth Southern African Conference for AI Research (SACAIR), 2023.
2. R. L. Myoya, F. Banda, V. Marivate, and **A. Modupe**, "Fine-Tuning Multilingual Pretrained African Language Models," 4th Workshop on African Natural Language Processing (AfricaNLP), 2023.

Preprints and Technical Reports

1. T. Sindane, U. Netshifhefhe, V. Marivate, **A. Modupe**, H. Combrink, and K. Sithole, "Maswao: An automatically collected, prepared, and benchmarked dataset for the South African Sign Language," arXiv:2611.02903 [cs.CL], 2026.
2. E. Tovurawa, B. Dlamini, S. Moore, V. Marivate, and **A. Modupe**, "Cacao Plant Disease Detection and Classification," Research

Square, 2025. doi:10.21203/rs.3.rs-5763786/v1.

3. W. Mapamela, B. Radiokan, S. E. Moore, **A. Modupe**, and V. Marivate, "Maize Disease Detection using Deep Learning Models," MIT Big Data Science Capstone (MIT808), University of Pretoria, 2024.
4. C. Gitau, S. Kabongo, **A. Modupe**, V. Marivate, T. Sindane, R. Lastrucci, and I. J. Dzingirai, "Masakhane Web: A Machine Translation Platform for African Languages," arXiv:2303.10666, 2023.
5. **A. Modupe**, T. Sindane, and V. Marivate, "Zero-Shot Transfer Learning using Affix and Correlated Cross-Lingual Embeddings," Authorea Preprints, 2023.

Awards and Honors

- Google Research Travel Grant, IEEE SMC 2023 – USD 3,000 (2023).
- IEEE SMC Conference Travel Grant – USD 500 (2023).
- SIT Research Grant Award, shared third place, University of Pretoria – ZAR 45,000 (2023).
- DSI–CSIR Inter-Bursary Support Programme – ZAR 360,000 (2018–2023).
- CSIR/DSI Inter-Bursary Support Programme – ZAR 96,000 (2009–2011).
- Black in AI Travel Grants, Black in AI Workshop at NeurIPS (2017–2019).
- Best Student Paper Award, International Conference of Information Security and Internet Engineering (ICSIE), World Congress on Engineering, London (2013).

Invited Talks and Presentations

- "All You Need Is Community," AI in Research Symposium, MIDAS, University of Michigan, Ann Arbor, MI, March 2025.
- "Foundations, Theory and Practice of E-Voting," Electoral Commission of South Africa, Cape Town, March 2025.
- AfriDSAI Teaching and Research Workshop on Data Science and AI, University of Pretoria, May 2025.
- "Responsible AI and Capacity Building," AI4D Partners Convening, University of Pretoria, October 2024.
- Session Moderator, Global Humanities Institute 2024: Design Justice AI (CHCI), University of Pretoria, June–July 2024.
- Conference presentations: IEEE SMC, Honolulu (2023); IEEE ICTAS, Durban (2017); WCE, London (2013).

Academic and Professional Service

University Service – University of Pretoria

- EBIT Faculty Postgraduate Committee (2023–2025).
- MIT Big Data Science Selection Committee (2022–2025).
- Computer Science Research Lab Technical Committee (2024–2025).
- DSFSI Research Lab Committee (2020–2025).
- External examiner for postgraduate dissertations, including at the University of the Witwatersrand, University of Limpopo, and University of Cape Town.

Professional Service

- Program Committee: IEEE BigData (2026); SACAIR (2023–present); SAICSIT (2022, 2024–present).
- Reviewer: ICLR (2024); ACM CIKM (2024); LoResLM (2025); International Journal of Intelligent Systems (2026); Discover Artificial Intelligence (2024); Deep Learning Indaba Publication Committee (2023).
- Mentor, Black in AI Emerging Leaders in AI (2025); member of the Masakhane NLP community.
- Professional memberships: IEEE, ACM, and ACL.

Industry Experience

Dupson LLC (AI consultancy) **2026–present**
Richmond, TX

Systems Engineer **2004–2008**
First City Monument Bank (FCMB), Lagos, Nigeria

Software Developer **1999–2004**
Alpha-Beta Consulting, Lagos, Nigeria

References

Available upon request.